Non-linear Dependence of Eddy-Current Damping on Magnet–Rotor Overlap Area

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RESULTS

The brake consists of two independent magnets assemblies that must be moved together. We define the overlapped area of the magnets and the copper rotor as $A_{\mathrm{ov}} = 2w\ell$, where $w = 1.80\,\mathrm{cm}$ is the width of the magnets, and ℓ is the insertion length of each assembly, and the factor of two accounts for the two assemblies. Eight settings were used, with ℓ from 0 to 2.1 cm in steps of 0.3 cm, corresponding to $A_{\mathrm{ov}} = 0$ – $7.56\,\mathrm{cm}^2$. The uncertainties $\sigma_w = 0.05\,\mathrm{cm}$ and $\sigma_\ell = \sqrt{0.05^2 + 0.02^2} \approx 0.054$ cm, the latter combines the ruler reading uncertainty and the paper-slip marking uncertainty, which gives $\sigma_A = 0.19$–$0.29\,\mathrm{cm}^2$.

The angular transducer was calibrated against the mechanical scale at seven different angles between −0.9 and 0.9 rad. The voltage is linear in angle with slope $a = 1.862\,\mathrm{V/rad}$, and the residuals of the linear fit are at most 0.054 rad. All angles were obtained as $\theta = V/a$ because the constant term given by the quotient of intercept and slope will absorbed by θ_{off}. In each run the rotor was released from rest at the amplitude of 0.5 rad from equilibrium, and the voltage was sampled at 50 Hz.

Each fit starts at the first sample lying 0.05 rad below the angle at which the rotor was held, so the hand-held segment before release is excluded from the overall data. For $\ell \le 1.2\,\mathrm{cm}$ the motion is oscillatory; these runs were fitted with Eq. (1) of Ref. [1], and Q was obtained from Eq. (3). For $\ell \ge 1.5\,\mathrm{cm}$ the rotor returns to equilibrium without oscillating (Fig. 1). These runs were fitted with the overdamped solution, Eq. (9), with ω_0 fixed at 5.495 rad/s, the mean of the underdamped runs.

The residuals of all ring-down fits show no slow drift and no structure that follows the rotor oscillation, so the fits well describe the main motion of the rotor.

The results are summarized in Table I below. The standard deviation s of the repeated releases is taken as the uncertainty of a single ring-down, and the standard error $s/\sqrt{n}$ as the uncertainty of each mean. At $\ell = 1.5\,\mathrm{cm}$, all three overdamped fits go with $\alpha = \sqrt{\beta^2 - \omega_0^2}$ [Eq. (8)] at its allowed lowest limit (i.e. $\alpha \to 0$, giving $Q = 0.500$. This setting is therefore at critical damping within the resolution of the fit, and the zero spread of the three values does not represent its true uncertainty.

Figure 2 shows the primary result. Q decreases monotonically with A_{ov}, from 166.2 ± 0.3 at $A_{\mathrm{ov}} = 0$ to 0.3195 ± 0.0004 at $7.56\,\mathrm{cm}^2$. The largest change occurs in the first step, where Q drops from 166.2 to 5.22.

As an independent check, Q was also obtained from the logarithmic decrement, Eqs. (10) and (11) of Ref. [1], for the two most oscillatory settings. The amplitude of each cycle was taken as half the peak-to-trough difference, and amplitudes $N = 20$ cycles apart ($\ell = 0$) or $N = 2$ cycles apart ($\ell = 0.3\,\mathrm{cm}$) were compared. This gives $Q = 163.6$–177.0 for $\ell = 0$, compared with 165.4–166.9 from the fits, and $Q = 5.41$–5.49 for $\ell = 0.3\,\mathrm{cm}$, compared with 5.19–5.27 from the fits.

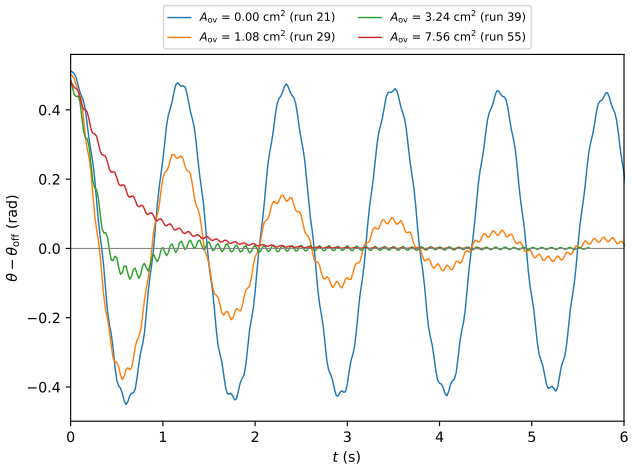


FIG. 1. Representative ring-downs, one for each of four overlap settings: $A_{\mathrm{ov}} = 0$, 1.08, 3.24 and $7.56\,\mathrm{cm}^2$ (runs 21, 29, 39 and 55). The angle from equilibrium, $\theta - \theta_{\mathrm{off}}$, is plotted against the time after release. As the overlap increases, the motion changes from a weakly damped oscillation to a return to equilibrium without oscillation. Angles were converted from the transducer voltage using the calibration slope 1.862 V/rad.

TABLE I. Quality factor at each overlap setting. n is the number of accepted ring-downs, $\bar{Q}$ the mean of the quilified factor, s the standard deviation of the single-run values, and SEM $= s/\sqrt{n}$ the standard error of the mean.

ℓ (cm)	A_{ov} (cm^2)	n	$\bar{Q}$	s	SEM
0.0	0.00	4	166.2	0.6	0.3
0.3	1.08	4	5.22	0.03	0.02
0.6	2.16	4	1.385	0.013	0.007
0.9	3.24	3	0.956	0.006	0.003
1.2	4.32	3	0.754	0.002	0.001
1.5	5.40	3	0.500[a]	—	—
1.8	6.48	3	0.4033	0.0008	0.0005
2.1	7.56	3	0.3195	0.0007	0.0004

[a] All three fits reached the critical-damping limit, $\alpha \to 0$.

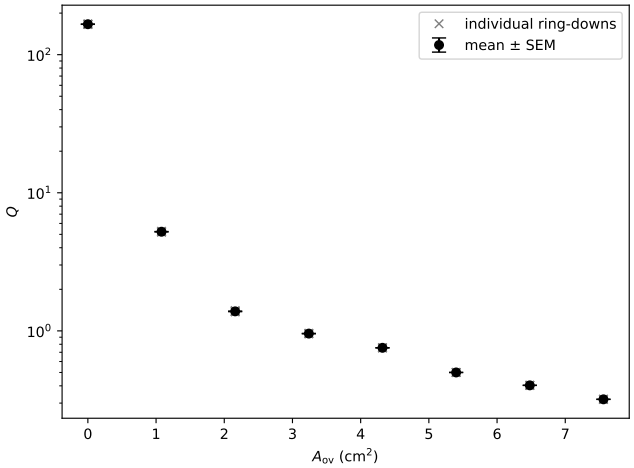


FIG. 2. Quality factor Q versus overlap area A_{ov}, on a logarithmic Q axis. Circles show the mean of the $n = 3$–4 ring-downs at each setting, with the standard error of the mean as error bars. The individual values (crosses) and the error bars are hidden behind the circles, because the spread between repeated releases is smaller than the symbol size; the individual spreads are listed in Table I. Q falls from 166.2 ± 0.3 at $A_{ov} = 0$ to 0.3195 ± 0.0004 at $7.56\,\mathrm{cm}^2$. Points with $A_{ov} \geq 5.40\,\mathrm{cm}^2$ were obtained with ω_0 fixed at 5.495 rad/s; at $5.40\,\mathrm{cm}^2$ all three fits reached the critical-damping limit, $Q = 0.500$.

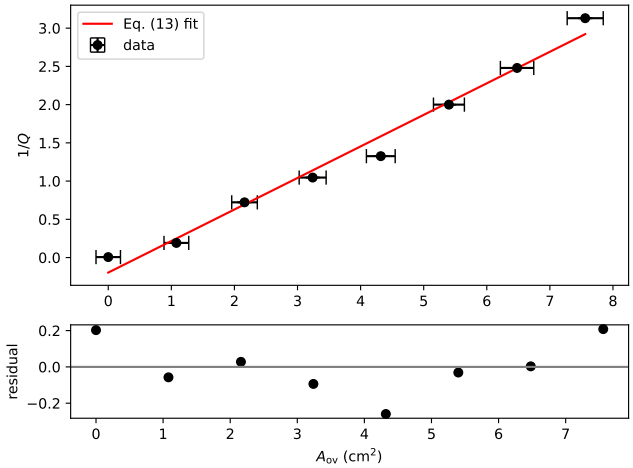


FIG. 3. Test of Eq. (13). Top: $1/\bar{Q}$ versus A_{ov}. The horizontal error bars, $\sigma_A = 0.19$–$0.29\,\mathrm{cm}^2$, come from the uncertainties in the pole-face width and the insertion length; the vertical error bars, the SEM propagated as σ_Q/Q^2, are smaller than the symbols. The line is an unweighted least-squares fit of Eq. (13), giving $1/Q_0 = -0.20 \pm 0.11$ and $k = 0.41(2)\,\mathrm{cm}^{-2}$. Bottom: residuals of the fit, positive at both ends of the range and negative near the middle.

To test the empirical model of Eq. (13), $1/\bar{Q}$ was fitted against A_{ov} by unweighted least squares (Fig. 3). The fit gives $1/Q_0 = -0.20 \pm 0.11$ and $k = 0.41(2)\,\mathrm{cm}^{-2}$. The residuals are not randomly scattered, where the largest positive residuals occur at the two ends of the range, and the most negative one near the middle, at $A_{ov} = 4.32\,\mathrm{cm}^2$. Even with the horizontal error bars included, the fitted line passes through the error bars of only about half of the points. A linear relation therefore does not describe $1/\bar{Q}$ over the full range of A_{ov}.

To check for drift, the magnet was returned to $\ell = 0.3$ cm at the end of the experiment (run 58). This revisit gives $Q = 3.95$, compared with the original mean of 5.22, a change of -24%. Its undamped angular frequency, $\omega_0 = 5.468$ rad/s, is close to the mean of 5.495 rad/s found for the underdamped runs.

DISCUSSION

A larger overlap places more of the moving copper in the magnetic field, so larger eddy currents are induced and more mechanical energy is dissipated in each cycle. The data agree with this qualitative picture: Q decreases monotonically with A_{ov}, and the motion changes from underdamped ($A_{ov} \leq 4.32\,\mathrm{cm}^2$) through critical damping near $5.40\,\mathrm{cm}^2$ to overdamped. The experiment establishes this monotonic dependence. It does not show that $1/Q$ grows in proportion to the overlap area, which is the assumption built into Eq. (13).

Two features show that Eq. (13) is not adequate over the measured range. First, the fitted intercept $1/Q_0 = -0.20 \pm 0.11$ is negative, which would imply negative damping, which is not physically possible, whereas the baseline ring-downs give a finite $Q = 166.2 \pm 0.3$. Second, the residuals follow a systematic curved pattern instead of scattering randomly about zero. The data therefore support a monotonic but non-linear dependence of $1/Q$ on A_{ov}.

The dominant systematic effect is the reproducibility of the magnet position. Repeated releases at a fixed position agree closely (at $\ell = 0.3$ cm, $s = 0.03$ for $\bar{Q} = 5.22$), but repositioning the magnet changed Q by -24%. Since ω_0 at the revisit is essentially unchanged, the restoring torque of the fiber did not change, and the difference must come from the magnet side: either an error in ℓ or a change in the air gap or alignment, which this experiment cannot distinguish. The horizontal error bars, $\sigma_A = 0.19$–$0.29\,\mathrm{cm}^2$, reflect the same limitation.

The second effect is the choice of fitting model at high overlap. For $\ell \geq 1.5$ cm, the values of Q rely on the assumption that ω_0 is the same as in the underdamped runs. At $\ell = 1.5$ cm, where the fits reach the critical limit, Q is determined only as $Q \approx 0.5$. The very small standard errors at 1.8 and 2.1 cm reflect the fixed ω_0, so the true uncertainty of these points is larger than the table suggests.

Other effects are minor. Release conditions and initial

amplitude have little influence, since the spread between repeated releases is small at every setting (Table I). The transducer calibration does not affect Q, because Eq. (3) depends only on the decay rate and the damped angular frequency, neither of which changes if the angle scale is rescaled; over the range used, the calibration is linear to within 0.054 rad.

The observed trend does not depend on the method used to extract Q. The logarithmic decrement gives values close to the fitted ones: the two ranges overlap at $\ell = 0$, and the decrement values are slightly higher at $\ell = 0.3\,\mathrm{cm}$. This difference is much smaller than the change in Q between neighbouring settings, and smaller than the change caused by repositioning the magnet.

CONCLUSION

Over $A_{\mathrm{ov}} = 0$–$7.56\,\mathrm{cm}^2$, the quality factor decreases monotonically from 166.2 ± 0.3 to 0.3195 ± 0.0004, passing through critical damping near $A_{\mathrm{ov}} = 5.40\,\mathrm{cm}^2$. The empirical model $1/Q = 1/Q_0 + kA_{\mathrm{ov}}$ is not supported over this range: the fit gives an unphysical intercept, $1/Q_0 = -0.20 \pm 0.11$, and its residuals follow a systematic pattern. The main limitation is the reproducibility of the magnet position, which changed Q by -24% when the setting $\ell = 0.3\,\mathrm{cm}$ was revisited. The strongest conclusion supported by the data is that the overlap area tunes the damping from weak damping to the overdamped regime, but $1/Q$ is not proportional to the overlap area.

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[1] PHYS 257 Experimental Methods I, *Lab 5: Damped Torsional Oscillator*, laboratory manual (McGill University, 2026).